

ASFI AHAMED

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Profile

Computer Science and Engineering undergraduate with hands-on experience building full-stack, AI-enabled, and data-driven software applications. Interested in developing practical software solutions to real-world problems.

Education

B.Sc. Engineering (Hons) in Computer Science & Engineering University of Moratuwa CGPA: 3.39/4.00 (through Semester 4)	2024 – Present Moratuwa, Sri Lanka
G.C.E. Advanced Level, KM/Al-Ashraq National School 3A Z-Score: 2.3250 Island Rank: 424	2022 Nintavur, Sri Lanka
G.C.E. Ordinary Level, KM/Al-Ashraq National School 9A	2019 Nintavur, Sri Lanka

Projects

NovaHub – AI-Enabled Collaboration Platform [Live App](#) | [GitHub](#) **2026 – Present**

Individual Project

- Built a multi-workspace collaboration platform with JWT authentication, workspace authorization, persistent messaging, secure invitation flows, and platform administration.
- Implemented real-time conversations with Socket.IO, including workspace rooms, reconnection handling, missed-message synchronization, and message read-state tracking.
- Developed AI workspace features for conversation summaries and evidence-grounded question answering using controlled backend tools, approved persistent memory, and an MCP-based read-only tool layer.

Tech: React, Node.js, Express, MongoDB, Socket.IO, MCP

Personal Portfolio & Custom CMS [Website](#) | [Demo](#) | [GitHub](#) **2026**

Individual Project

- Built a custom portfolio CMS with a protected admin dashboard for managing projects, skills, education, experience, certificates, social links, and site content without editing frontend source code.
- Implemented database-backed CRUD, search, filtering, ordering, validation, media management, authentication, storage, and row-level security for protected content management.

Tech: React, TypeScript, Supabase, PostgreSQL, Tailwind CSS

ViewCastLK – YouTube View Count Forecasting [Live App](#) | [GitHub](#) **2026 – Present**

Team Project – 3 Members

- Contributed to a multi-horizon ML system for forecasting future viewership trends of Sri Lankan YouTube content using pre-publication metadata and historical channel information.
- Built and refined the React forecasting dashboard and integrated it with a production FastAPI prediction service and model artifacts.
- Contributed to API-based data collection, PostgreSQL-backed workflows, containerized deployment, and CI/CD for the production system.

Tech: Python, React, FastAPI, PostgreSQL, REST APIs, Docker, Google Cloud Run

NoteBot – PDF RAG Study Assistant [Live App](#) | [GitHub](#) **2026**

Individual Project

- Built a PDF question-answering pipeline with page-aware text extraction, chunking, embeddings, similarity retrieval, and context-grounded LLM responses.
- Implemented cosine-similarity retrieval and automated tests for retrieval and AI-provider behavior, keeping retrieved document evidence separate from generated responses.

Tech: Python, Streamlit, RAG, Embeddings, LLM APIs

Student Performance Predictor [Live App](#) | [GitHub](#) **2026**

Individual Project

- Built an end-to-end regression pipeline covering missing-value handling, categorical encoding, feature scaling, model comparison, tuning, persisted artifacts, and production inference.
- Compared eight regression algorithms, tuned seven using 3-fold GridSearchCV, and selected Linear Regression with a test-set R^2 of 0.8804.
- Exposed the trained preprocessing and prediction pipeline through a Flask application deployed on AWS Elastic Beanstalk.

Tech: Python, Pandas, NumPy, scikit-learn, Flask, AWS Elastic Beanstalk

Individual Project

- Developed an item-based collaborative-filtering system using explicit user ratings, cosine distance, and a sparse book-by-user interaction matrix.
- Built the preprocessing, model-artifact, and inference workflow and packaged recommendations into an interactive Streamlit application.

Tech: Python, Pandas, SciPy, scikit-learn, Streamlit

Research & Publication**Environmental Drivers of Respiratory Disease: A District-Level Analysis**[Paper](#) | [GitHub](#) 2026

Accepted at MERCon 2026

- Analyzed multi-source environmental and respiratory-health data across all 25 districts of Sri Lanka to study relationships between environmental conditions and respiratory disease patterns.
- Developed temporally validated XGBoost models to evaluate predictive relationships while reducing the risk of temporal data leakage.
- Applied SHAP-based model interpretation to identify and explain the environmental factors contributing most strongly to model predictions.

Technical Skills

Languages: JavaScript, Python, SQL

Frontend: React, HTML, CSS, Tailwind CSS

Backend & APIs: Node.js, Express, REST APIs, Socket.IO, FastAPI, Flask

Databases: PostgreSQL, MongoDB, Supabase, MySQL

AI & Machine Learning: RAG, Embeddings, LLM API Integration, scikit-learn, XGBoost, SHAP

Data: Pandas, NumPy, SciPy

Tools: Git, GitHub, Docker, GitHub Actions

Certifications

AWS Academy: Cloud Foundations; Microservices and CI/CD Pipeline Builder (2026)

Kaggle Learn: Python, Pandas, Intro to Machine Learning, Data Cleaning, Feature Engineering (2026)

NVIDIA: Fundamentals of Accelerated Data Science (2025)

Generative AI: Generative AI for Beginners – Build AI Chatbot (2026)

Activities

Company Coordinator – Career Fair Day, University of Moratuwa (2026) | **OC Logistics Member** – SLIoT | **Logistics Support** – Sakura Event

Reference

Prof. Dulani Meedeniya

Professor, Department of Computer Science & Engineering

University of Moratuwa, Sri Lanka

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